

Cosc 30: Discrete Mathematics

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Give good estimates to the following functions of n , asymptotically tight under Θ notation.

Problem 1.

$$\left(\sum_{i=0}^{30} n^i \right)^{30}$$

Solution. $\Theta(n^{900})$

Problem 2. The running time of the following program:

```
STOOGESORT(A[0 .. n-1]):  
    if n == 2 and A[0] > A[1]:  
        swap A[0] <-> A[1]  
    else if n > 2:  
        m <- ceil(2n/3)  
        STOOGESORT(A[0 .. m-1])  
        STOOGESORT(A[n-m .. n-1])  
        STOOGESORT(A[0 .. m-1])
```

Solution. $\Theta(n^{\log_{3/2} 3})$

Problem 3. The number of array entries printed by the following program:

```
HPRINT(A[0 .. n-1]):  
    for k from 1 to n:  
        ptr <- 0  
        while ptr < n:  
            print A[ptr]  
            ptr <- ptr + k
```

Solution. $\Theta(n(\ln(n)))$

Problem 4. Give a tight asymptotic estimate for

$$\prod_{k=1}^n e^{1/k}.$$

Solution. $\Theta(n)$

Problem ★5. Give a tight asymptotic estimate for

$$\sum_{k=1}^n k^{30} \cdot \left(\frac{1}{30}\right)^k.$$

Solution. $\Theta(1)$

Problem ★6. Give a tight asymptotic estimate for

$$\sum_{k=1}^n \frac{1}{k \log k \log \log k}.$$

Solution. $\Theta(\log \log \log n)$